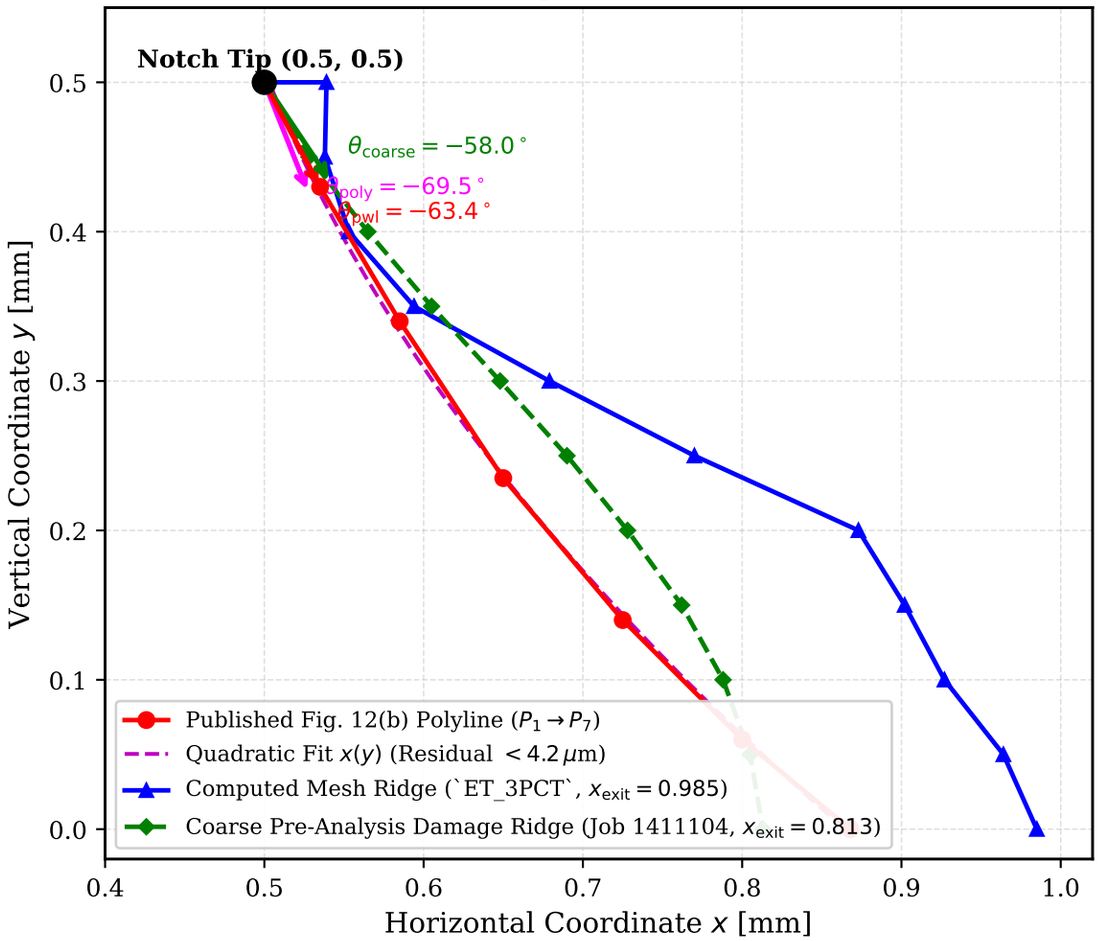
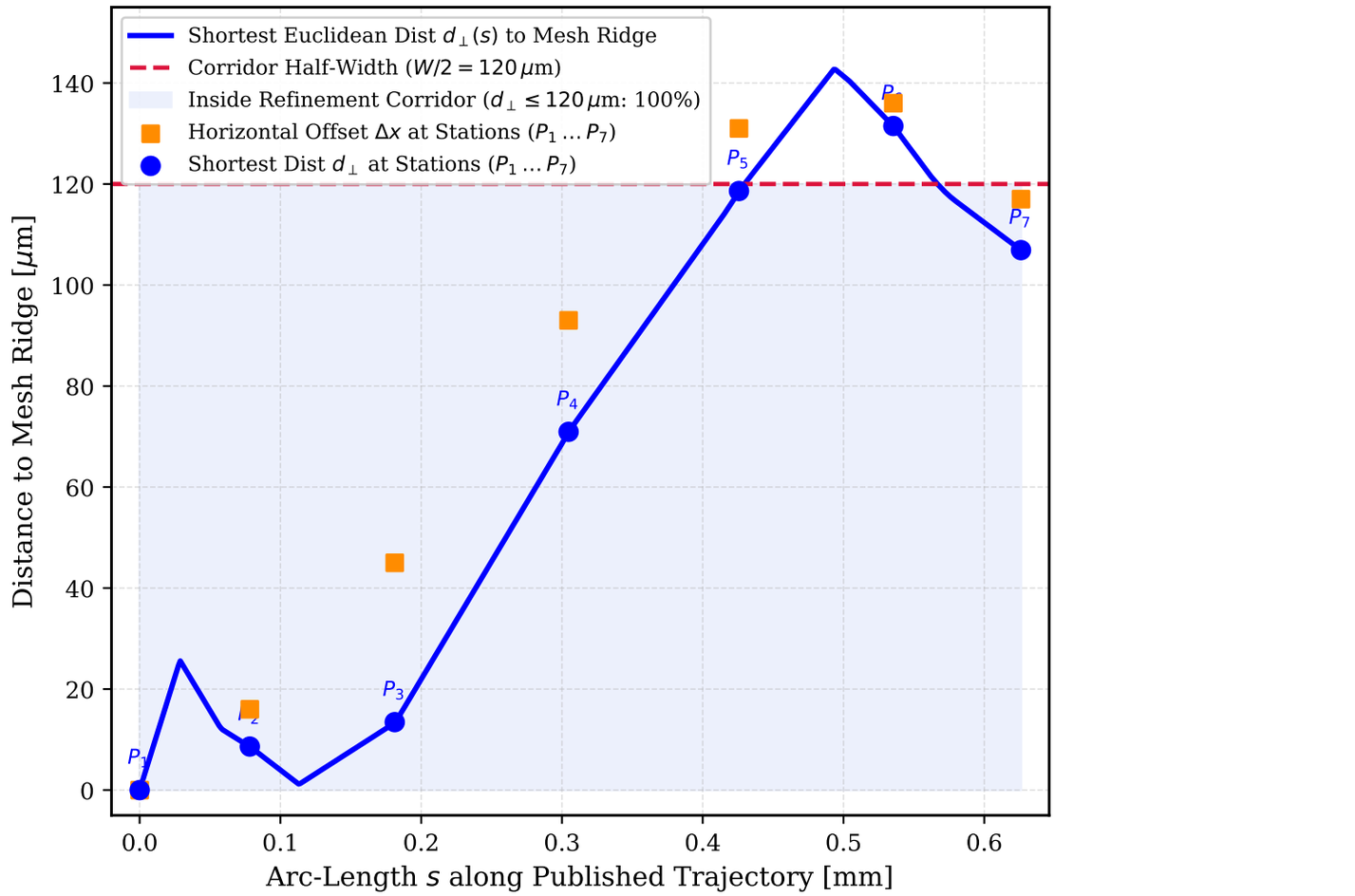
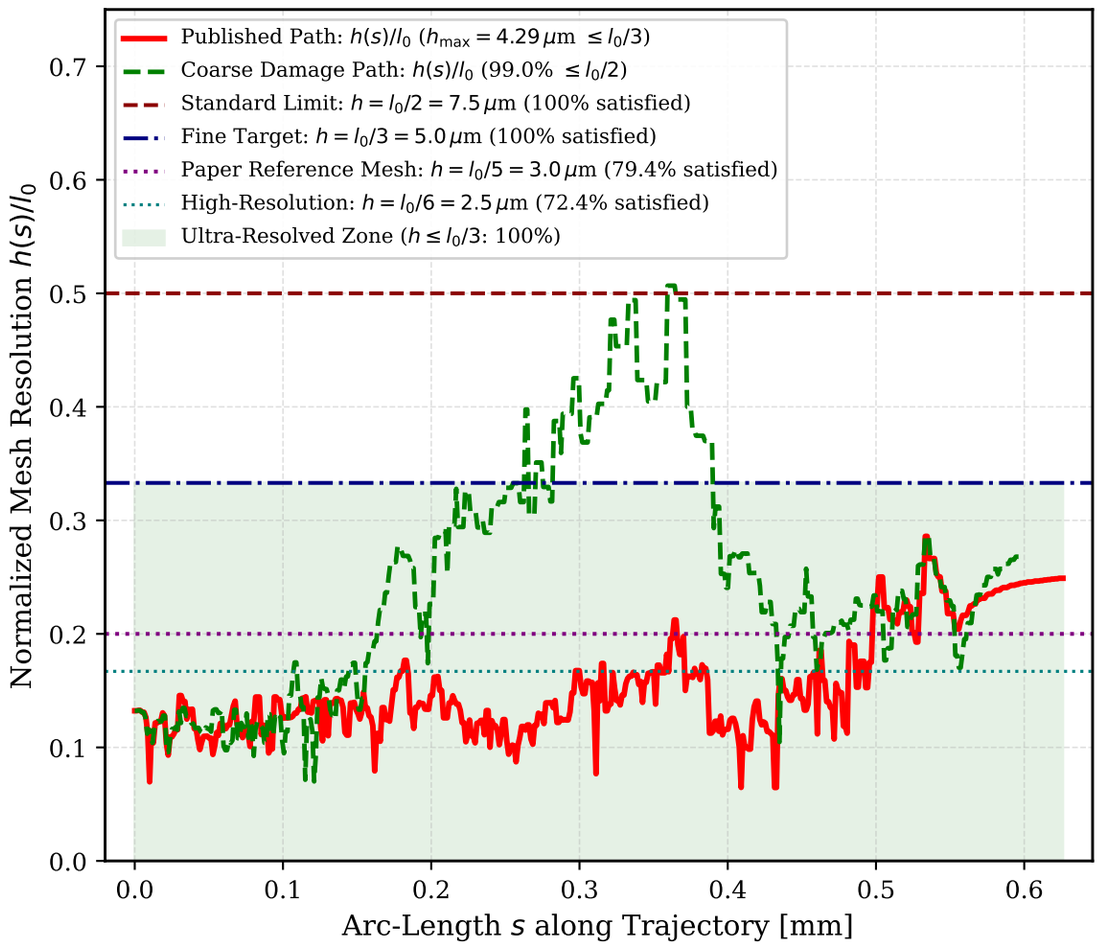
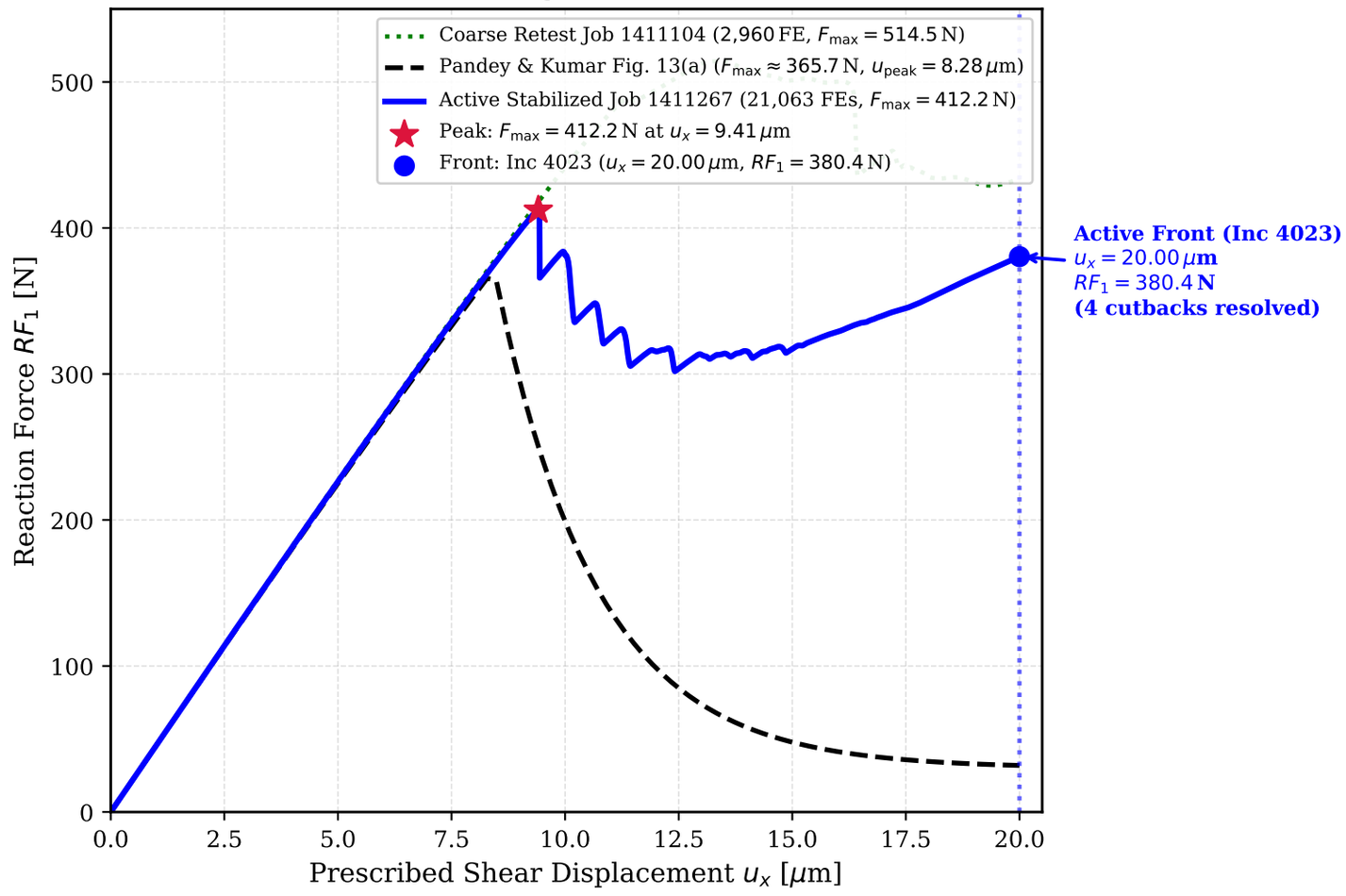

(a) Mode-II Crack Trajectories & Initiation Vectors
Notch Tip (0.5, 0.5)
$\theta_{coarse} = -58.0^\circ$
$\theta_{poly} = -69.5^\circ$
$\theta_{owl} = -63.4^\circ$
Published Fig. 12(b) Polyline ($P_1 \to P_7$)
Quadratic Fit $x(y)$ (Residual $< 4.2\,\mu$m)
Computed Mesh Ridge (`ET_3PCT`, $x_{exit} = 0.985$)
Coarse Pre-Analysis Damage Ridge (Job 1411104, $x_{exit} = 0.813$)
Vertical Coordinate y [mm]
Horizontal Coordinate x [mm]


(b) Horizontal Offset Δx vs. Shortest Distance $d_\perp$
Shortest Euclidean Dist $d_\perp(s)$ to Mesh Ridge
Corridor Half-Width ($W/2 = 120\,\mu$m)
Inside Refinement Corridor ($d_\perp \le 120\,\mu$m: 100%)
Horizontal Offset Δx at Stations ($P_1 \dots P_7$)
Shortest Dist $d_\perp$ at Stations ($P_1 \dots P_7$)
P_1
P_2
P_3
P_4
P_5
P_6
P_7
Distance to Mesh Ridge [μm]
Arc-Length s along Published Trajectory [mm]


(c) Local Element Resolution along Crack Paths ($l_0 = 15.0\,\mu$m)
Published Path: $h(s)/l_0$ ($h_{max} = 4.29\,\mu$m $\le l_0/3$)
Coarse Damage Path: $h(s)/l_0$ (99.0% $\le l_0/2$)
Standard Limit: $h = l_0/2 = 7.5\,\mu$m (100% satisfied)
Fine Target: $h = l_0/3 = 5.0\,\mu$m (100% satisfied)
Paper Reference Mesh: $h = l_0/5 = 3.0\,\mu$m (79.4% satisfied)
High-Resolution: $h = l_0/6 = 2.5\,\mu$m (72.4% satisfied)
Ultra-Resolved Zone ($h \le l_0/3$: 100%)
Normalized Mesh Resolution $h(s)/l_0$
Arc-Length s along Trajectory [mm]


(d) Production Solve (Job 1411267) vs Baselines
Coarse Retest Job 1411104 (2,960 FE, $F_{max} = 514.5$ N)
Pandey & Kumar Fig. 13(a) ($F_{max} \approx 365.7$ N, $u_{peak} = 8.28\,\mu$m)
Active Stabilized Job 1411267 (21,063 FEs, $F_{max} = 412.2$ N)
Peak: $F_{max} = 412.2$ N at $u_x = 9.41\,\mu$m
Front: Inc 4023 ($u_x = 20.00\,\mu$m, $RF_1 = 380.4$ N)
Active Front (Inc 4023)
$u_x = 20.00\,\mu$m
$RF_1 = 380.4$ N
(4 cutbacks resolved)
Reaction Force RF_1 [N]
Prescribed Shear Displacement u_x [μm]